# Initial Analytical Application

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## Loading packages

```
library(tidyverse)
```

## Import dataset

```
All_Country <- read_csv("Data/Processed/AllCountries_Processed.csv")
```

## Analysis

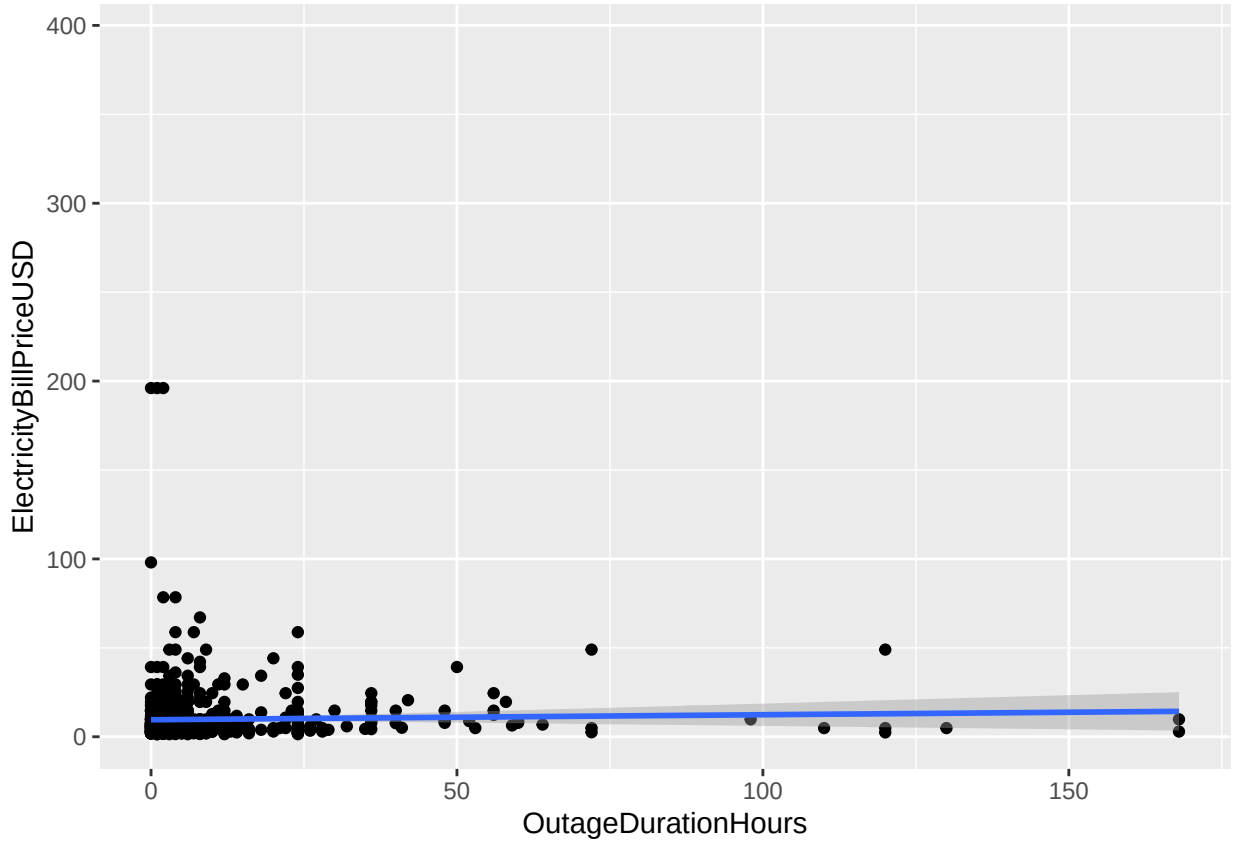
Linear regression model between “Electricity Bill Price” and “Outage Duration Hours” to see if the higher the electricity bill, the more stable the connection is.

```
#Kenya
All_Country %>%
  filter(Country == "Kenya") %>%
  ggplot(aes(x = OutageDurationHours, y = ElectricityBillPriceUSD)) +
  geom_point() +
  geom_smooth(method = "lm")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 3867 rows containing non-finite outside the scale range
## (`stat_smooth()`).
```

```
## Warning: Removed 3867 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

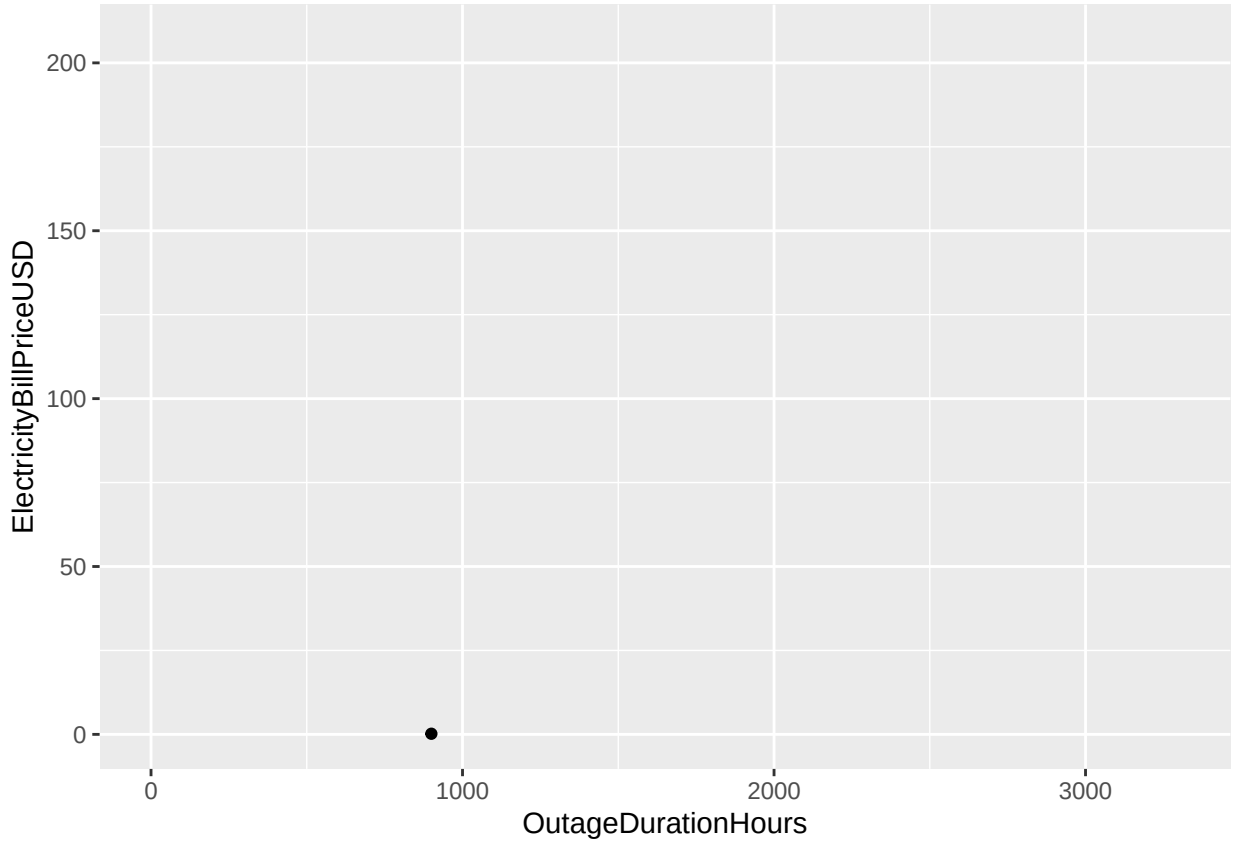



```
#Liberia
All_Country %>%
  filter(Country == "Liberia") %>%
  ggplot(aes(x = OutageDurationHours, y = ElectricityBillPriceUSD)) +
  geom_point() +
  geom_smooth(method = "lm")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 3902 rows containing non-finite outside the scale range
## (`stat_smooth()`).
```

```
## Warning: Removed 3902 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

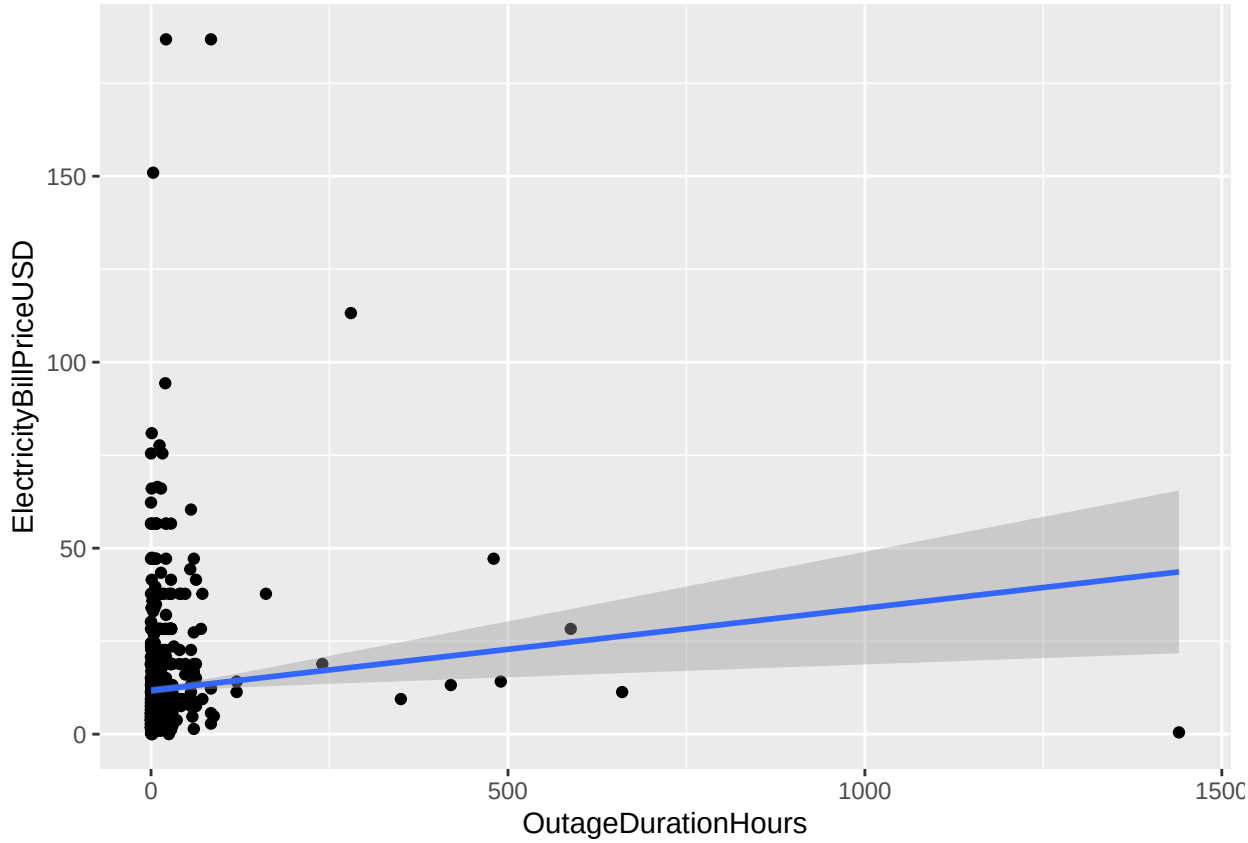



```
#Niger
All_Country %>%
  filter(Country == "Niger") %>%
  ggplot(aes(x = OutageDurationHours, y = ElectricityBillPriceUSD)) +
  geom_point() +
  geom_smooth(method = "lm")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 3076 rows containing non-finite outside the scale range
## (`stat_smooth()`).
```

```
## Warning: Removed 3076 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

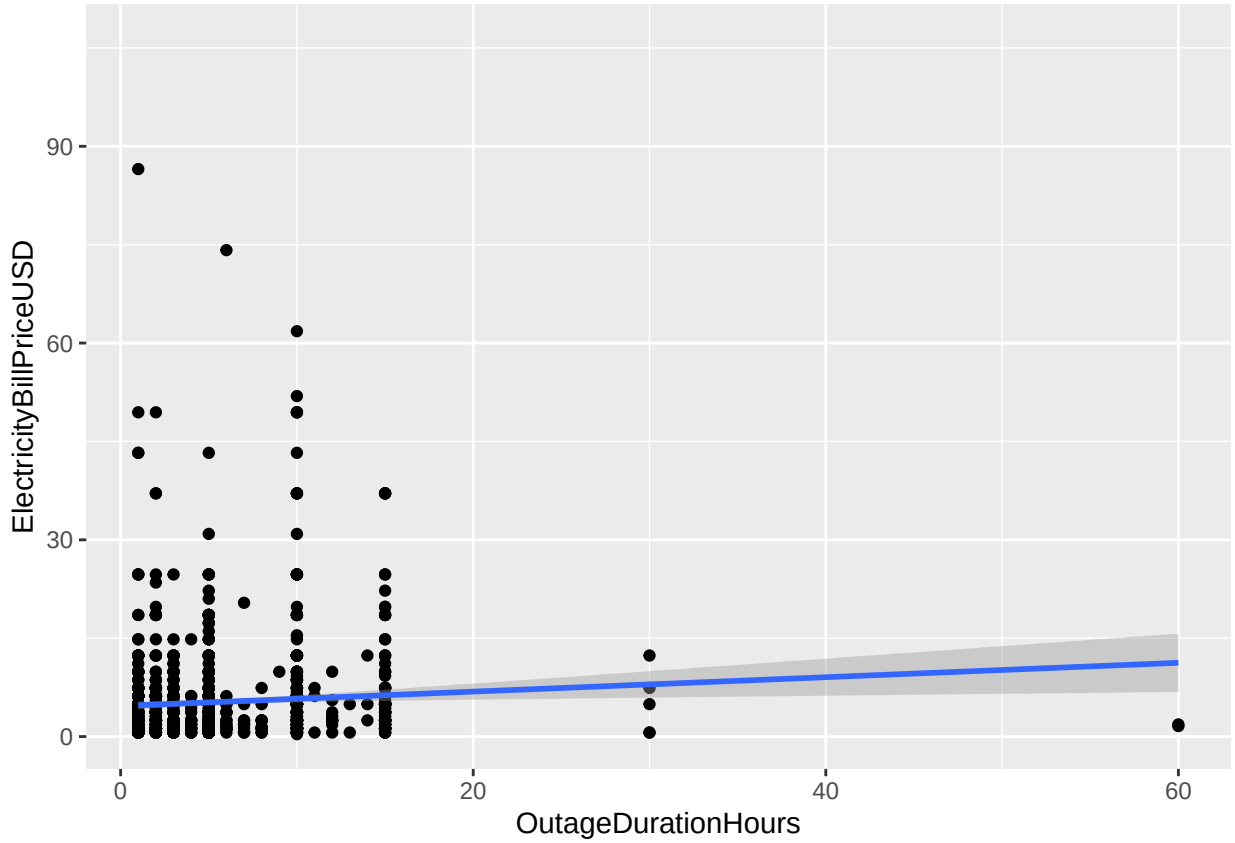



```
#Rwanda
All_Country %>%
  filter(Country == "Rwanda") %>%
  ggplot(aes(x = OutageDurationHours, y = ElectricityBillPriceUSD)) +
  geom_point() +
  geom_smooth(method = "lm")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 2031 rows containing non-finite outside the scale range
## (`stat_smooth()`).
```

```
## Warning: Removed 2031 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

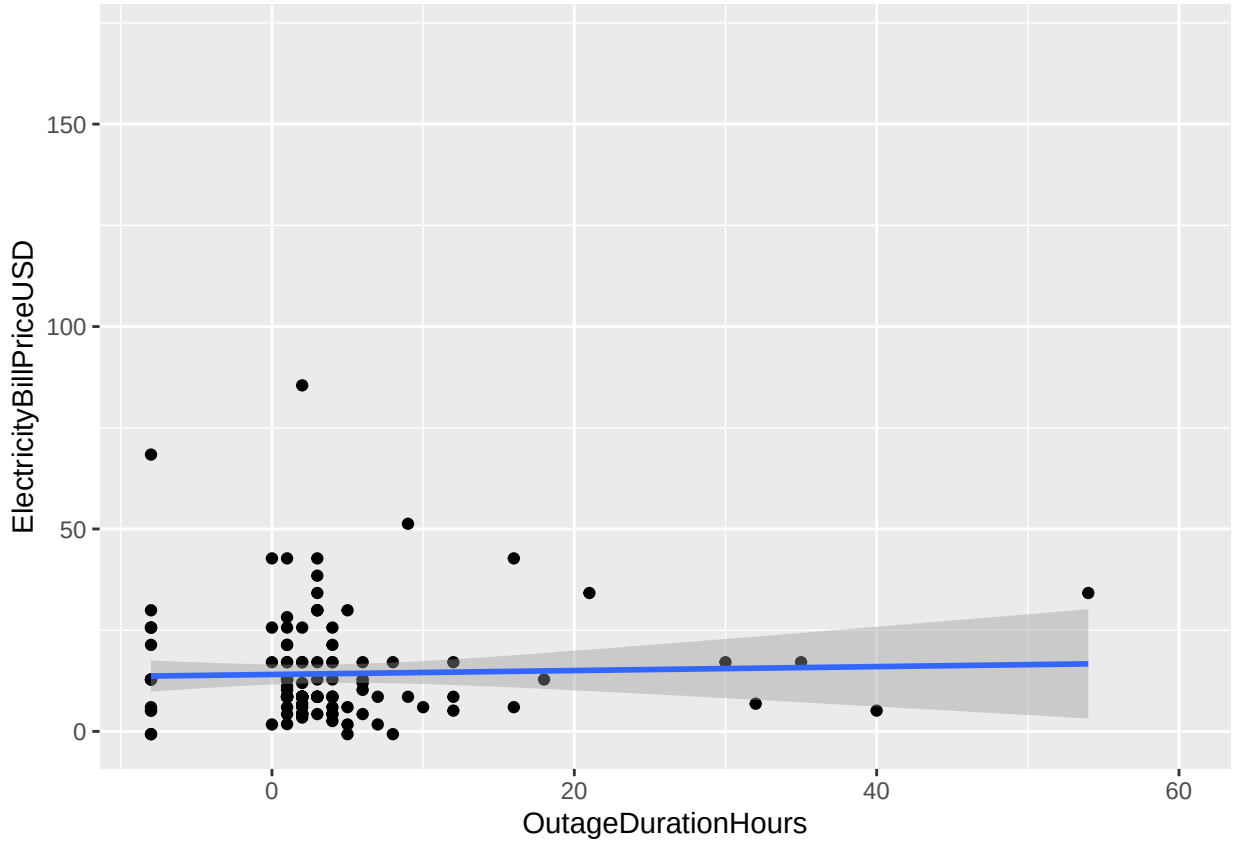



```
#Zambia
All_Country %>%
  filter(Country == "Zambia") %>%
  ggplot(aes(x = OutageDurationHours, y = ElectricityBillPriceUSD)) +
  geom_point() +
  geom_smooth(method = "lm")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 3483 rows containing non-finite outside the scale range
## (`stat_smooth()`).
```

```
## Warning: Removed 3483 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



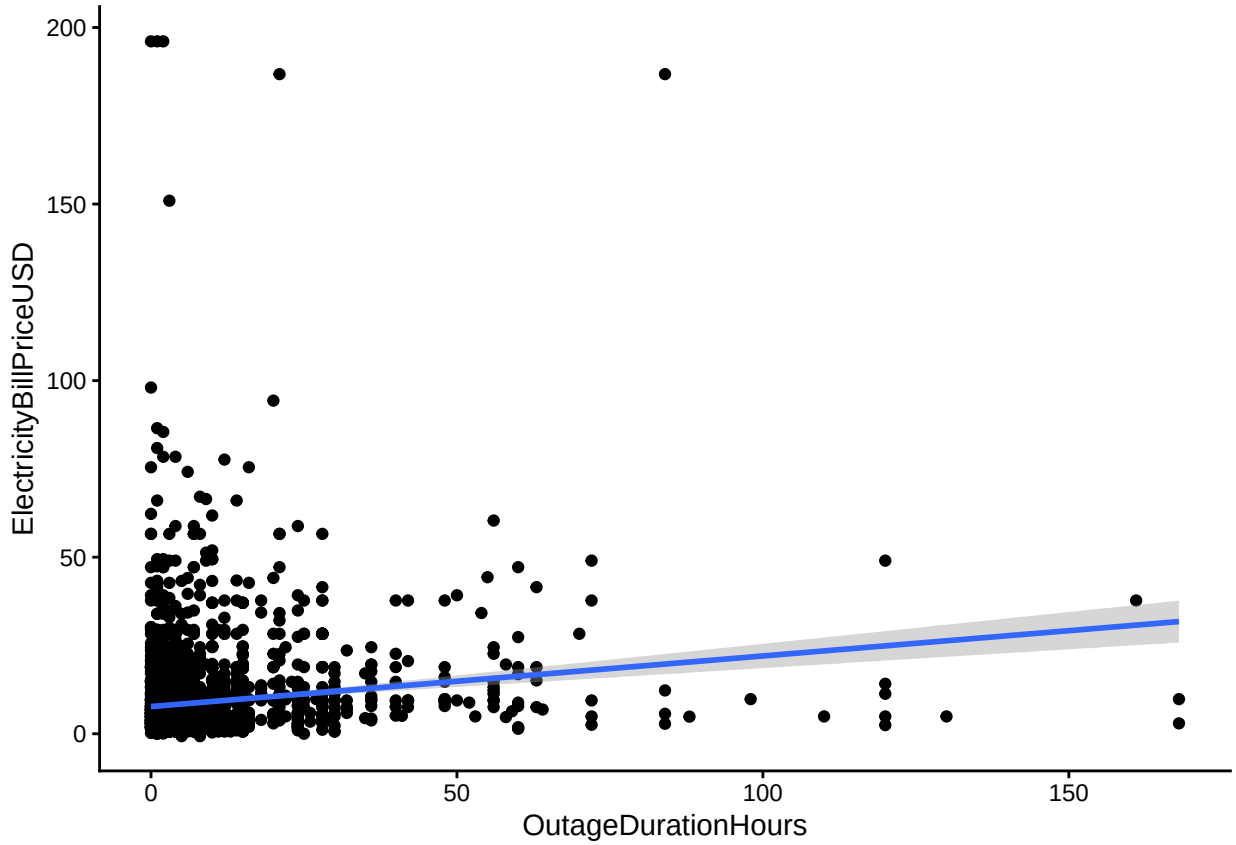

Liberia is currently missing the values on Electricity bill price. Niger has some very large values on outage duration hours. Zambia somehow has some negative values on outage duration hours.

```
All_Country %>%
  filter(OutageDurationHours >= 0 & OutageDurationHours <= 168) %>%
  ggplot(aes(x = OutageDurationHours, y = ElectricityBillPriceUSD)) +
  geom_point() +
  geom_smooth(method = "lm") +
  theme_classic()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 960 rows containing non-finite outside the scale range
## (`stat_smooth()`).
```

```
## Warning: Removed 960 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




I removed the values larger than 168 hours in the outage duration hours column, because a week can only have 168 hours and any number larger than that is likely an error.

```
All_Country_Filtered <- All_Country %>%
  filter(OutageDurationHours >= 0 & OutageDurationHours <= 168)

LinearRegression <- lm(ElectricityBillPriceUSD ~ OutageDurationHours,
                       data = All_Country_Filtered)

summary(LinearRegression)
```

```
##
## Call:
## lm(formula = ElectricityBillPriceUSD ~ OutageDurationHours, data = All_Country_Filtered)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -28.825  -5.957  -3.473   1.037 188.378
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)          7.70078    0.27064  28.454  < 2e-16 ***
## OutageDurationHours  0.14325    0.01879   7.625 3.23e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 12.86 on 3064 degrees of freedom
##   (960 observations deleted due to missingness)
## Multiple R-squared:  0.01862,    Adjusted R-squared:  0.0183
## F-statistic: 58.15 on 1 and 3064 DF,  p-value: 3.226e-14
```

From the Linear Regression summary the R value is 0.01862, showing a very week positive correlation between Electricity Bill Price and Outage Duration Hours.